

A Network-Centric Architecture for Building the Cloud Continuum

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Abstract—The growing interest in distributed, context-aware and data-sensitive applications is pushing the evolution of computing infrastructures from centralized to distributed models, which could effectively tackle the execution of complex software frameworks over geographical scale. The concept of *cloud continuum* that extends computing infrastructures beyond the data center boundary will require new architectural paradigms that overcome the evident limitations intrinsic in mere cloud federation and that enable effective and efficient interaction between the cloud and the physical environment.

In this paper, we discuss why and how telecommunication networks could be the most effective infrastructure to create a distributed, pervasive, carrier-grade cloud continuum, by acting as the core federation paradigm for the dynamic and flexible composition of data centers, networks and IoT platforms.

I. INTRODUCTION

The Infrastructure-as-a-Service (IaaS) cloud paradigm provides virtual environments to run applications and services, abstracting most issues concerning the underlying infrastructure. IaaS systems usually virtualizes a single or a few huge data centers; this greatly facilitates the operation and management of the physical infrastructure, but makes them “far” from users, physical devices, and other similar installations, hence making hard an effective and efficient interaction between the cloud and the physical environment. This is a serious challenge for distributed, context-aware and data-sensitive applications, like most cyber-physical systems (interactive voice/video, gaming, smart metering, car pilot, grid protection, high frequency trading, etc.), that require proximity to the users in order to meet very stringent performance requirements (mainly in terms of latency and available bandwidth) [1].

The need for closer *proximity* has pushed the realization of large scale, robust and redundant computing environments as *distributed clouds*, where different data centers are operated under a common IaaS interface by a single infrastructure provider. This model works for coarse-grained localization (e.g., by deploying a different data center for each continent or world macro-area, as already done by some big cloud providers), but is not economically sustainable for a capillary worldwide coverage. The notion of *cloud federation* envisions a collection of heterogeneous infrastructures, by pooling

physical resources from multiple providers, which potentially span the entire globe; however, it brings several issues concerning the management of a multi-domain, multi-tenant, heterogeneous, distributed infrastructure: deployment, networking, run-time management, orchestration, security and privacy [2].

The concept of *fog computing* has been proposed to extend the cloud paradigm at the very edge of the network [3], building on the increasing availability of computing and storage resources in the access segment [4]. Though such installations could be seen as micro data centers and network virtualization would enable to seamless interconnect them, the undergoing evolution towards 5G infrastructures is promising a radical transformation of communication infrastructures, well beyond the mere *Network-as-a-Service* paradigm.

Indeed, future access networks are expected to integrate far more elements in addition to transmission lines and switching equipment, “from the Things to the Cloud” [5]: computing, storage, things, network functions, hardware accelerators. The network will represent the nervous system that interconnects all such elements with data centers and Internet-connected smart devices (things) [4]. This large collection will basically become a rich ICT programmable fabric, raising the opportunity for creating a seamless and pervasive landscape of clouds, pictorially resembling a “cloudscape”.

According to the above vision, we believe that 5G networks will be able to provide distributed IaaS services over geographical scale, including far more abstractions than today (compute, storage, connectivity, things, network functions, etc.), with the traditionally carrier-grade service that have always characterized network operators, hence leading to network-centric cloud architectures. To this aim, a management and control framework is needed to abstract the underlying heterogeneous ICT fabric, including traditional data center installations, and to provide suitable service abstractions, turning into a pervasive cloud continuum that hides the complexity and heterogeneity of the underlying infrastructure and eases the design, development and deployment of distributed applications and services¹.

¹This vision is currently endorsed by innovation managers of telco operators [5].

In this paper, we outline the basic architectural components for such a control framework and review existing technologies that already implement part of the basic concepts. The paper is organized as follows. In Section II we review background and related work. Section III outlines our Cloudscape architecture, then in Section IV we discuss how applications and services benefit from our approach. Finally, we give our conclusions in Section V.

II. BACKGROUND AND RELATED WORK

Typical cloud federation schemes consider interoperability issues (resource discovery and management, identity management, authentication, Service Level Agreement management, application brokerage) among different clouds, but do not take into account integration with wide-area networks, which are traditionally seen just like the underlying communication facility rather than constituent part of the computing eco-system [6]–[8]. Indeed, networking plays a crucial role in distributed cloud computing: it is in charge of providing remote delivery of computing resources, and it was demonstrated that networking performance has a significant impact on the quality of Cloud services [9]. However, the set of virtualization technologies currently offered by telco operators are usually static in nature and cannot be automatically created, terminated and scaled in short time frames, hence failing to provide per-application, per-service and per-tenant explicit guarantees in a dynamic and heterogeneous environment [10].

The need for dynamic connectivity and guaranteed service levels has motivated the interest in network virtualization, which basically consist in decoupling service provisioning from network infrastructure and exposing underlying network functionalities through resource abstraction. A number of proposal have addressed the issue by introducing different kinds of controllers [11]; however, the disruptive evolution in network control and operation brings the potential for a radical change, from the traditional model where the network dictates communication capabilities, to a new paradigm where applications shape the network according to their requirements. In this respect, the “intent” framework designed within the Open Network Operating System (ONOS) [13] is one of first attempts in the direction of true *Application-Defined Networking*.

Service-Oriented Architecture (SOA) offers a promising approach to network virtualization, by effective, flexible, and scalable interaction among key players in a converged networking and Cloud computing environment [9]. SOA provides effective architectural principles for heterogeneous system integration, by encapsulating system resources and capabilities in the form of services and provides a loose-coupling interaction mechanism among these services.

However, the heterogeneity of the underlying ICT fabric (especially when computing, storage and things

are taken into account), the need for carrier-grade performance, and a flexible intent abstraction, require a more structured organization, where services are organized into functional layers that facilitate the design, deployment, integration and operation of large complex systems.

III. THE CLOUDSCAPE ARCHITECTURE

We argue that the rich set of heterogeneous resources and the large number of connected things naturally candidate telecommunication networks to become a computing continuum, offering Everything-as-a-Service (XaaS) facility for building pervasive and interactive applications [4]. Our belief stems from the following considerations:

- networks constitute the backbone of every distributed computing environment;
- distributed cloud abstractions are already being developed for communication networks, in order to run NFV and SFC frameworks;
- all elements (end devices, users, data centers) must have some form of subscription to the network;
- different telco operators can easily federate to provide broader coverage;
- telco infrastructures have been managing millions of subscribers for a long time;
- a distributed system is usually easier to manage with central coordination;
- mobility management has been integral part of network design since a long time;
- traditional reliability of communication infrastructures is very high, usually achieving five nines availability guarantees (99.999%).

Future networks will be able to manage heterogeneous ICT infrastructures and to build cloud services on top, giving the sensation of a pervasive, wide, seamless, carrier-grade cloud continuum, i.e., a *cloudscape* (see Fig. 1) [5]. To this aim, we outline the main layout of a Cloudscape Operating System (CLOS) for cloud federation, i.e., a modular software framework organized in the same way as a modern Operating System (OS) that *i)* aggregates and interconnects geographically dispersed infrastructures, *ii)* exposes virtual resources according to common abstraction models for the underlying hardware, *iii)* provides a set of utility “libraries” to facilitate common tasks (e.g., authentication and authorization, connectivity, access to context information, redundancy, ...), and *iv)* enables to deploy and run cognitive and pervasive applications over a uniform and consistent execution environment. This approach will provide a homogeneous abstraction of the underlying infrastructure spanning over multiple administrative domains and, most of all, will enable the composition of cloud and communication services with predictable and guaranteed network performance.

A modern OS typically includes hardware abstraction (drivers that provide common models and interfaces for

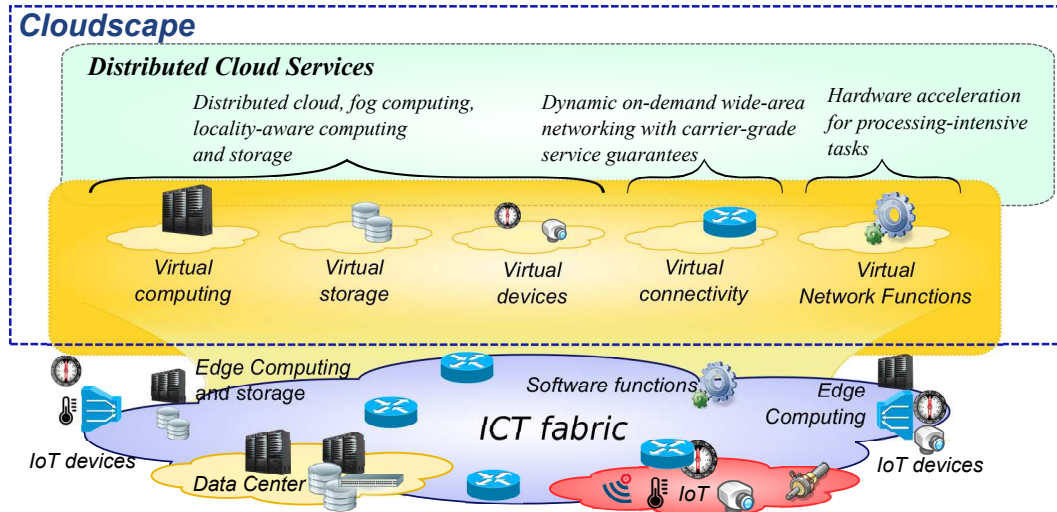


Fig. 1. The Cloudscape abstraction: the network is the Operating System of an heterogeneous ICT fabric made of communication equipment, data centers, fog installations, IoT.

accessing physical devices), core system functions (command execution, I/O, storage and processing control, networking functions, inter-process communications, hardware abstraction, resource monitoring), and userland services (programmatic interfaces and Application Program Interfaces – APIs – to system functions, like launching applications, browsing the file system, interpreting commands, etc.). On top of this structure, user applications run and exploit the OS facilities. Similarly, the CIOS architecture is made of three logical tiers, from the bottom up (see Figure 2):

- the **Southbound Interface**, which includes drivers to abstract ICT installations (data centres, networks, IoT) and to decouple control and management operations from the specific infrastructure;
- the system **Kernel**, which federates different domains and is responsible for resource abstraction (virtual CPU, virtual RAM, virtual networking, storage, IoT virtual devices), reliable and resilient operation, infrastructure configuration, authentication and authorization, and management and monitoring, targeting performance levels close to five nines (99.999%) telecommunication standards;
- the **Userland**, which exposes API and interfaces to common functions and services.

A. Southbound Interface

The Southbound Interface identifies functional models for each relevant ICT facility (e.g., Data Centre, Fog, Switching fabric, Thing), which include capability and functionality description. For each resource, the model will include identification, classification, description, capability, security, and programmatic interfaces. This facilitates common management tasks by the kernel, for instance migration of virtual resources, by hiding the jungle of different execution environments (Virtual Machines, Containers, Unikernels) and data representation

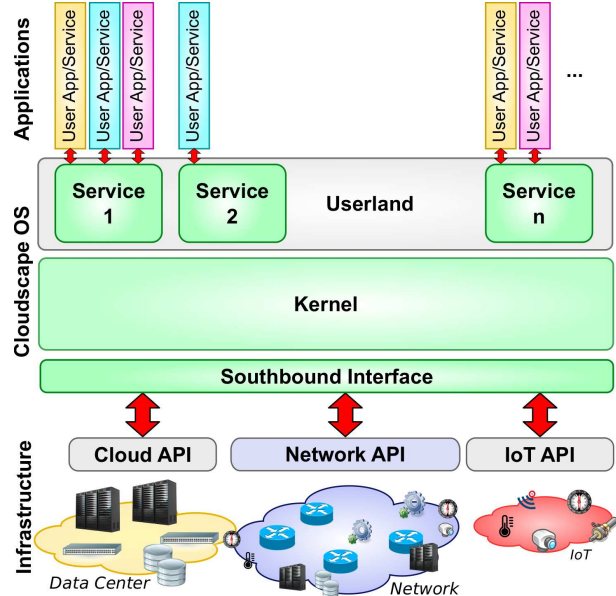


Fig. 2. Layout of the Cloudscape Operating System

formats that make application migration a hard task in a heterogeneous setting. The Southbound is composed of drivers, which translate the Cloudscape functional model (a set of common system calls) into the specific resource interface (e.g., OCCI², Amazon E2C³, NetConf [14], OpenFlow [15], SMTP [16], IoT platform [17], etc.). According to the envisioned approach, adding new infrastructures will be as simple as writing an additional *device driver*. The logical structure is shown in Fig. 3.

²Open Cloud Computing Interface (OCCI), delivered through the Open Grid Forum. URL: <http://occi-wg.org/>.

³Amazon Elastic Compute Cloud (EC2), <http://aws.amazon.com/ec2>

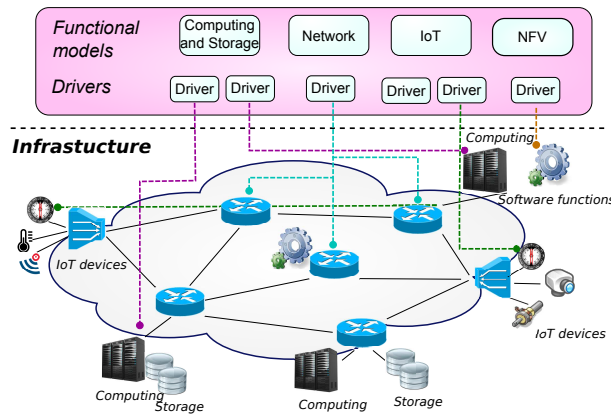


Fig. 3. The CIOS Southbound Interface.

B. Kernel layer

The Kernel layer accounts for the following aspects:

- discovery, registration, configuration, control and management of available programmable infrastructures;
- exposing virtual resources to upper layers;
- mapping virtual resources to physical infrastructures, according to locality, privacy, security, trustworthiness, and performance principles, ;
- support for placement policies (e.g., energy-efficiency, load balancing, resource utilization, security), including the seamless and transparent migration of virtual resources in case of hardware failure;
- dependable, resilient, scalable, and highly-available operation, through a design that accounts for a symmetric collection of instances that provide a resilient, distributed and horizontally scalable solution (see Fig. 4);
- real-time resource monitoring and data analytics, which will proactively detect, infer and forecast any misbehavior, risk of failure, deviation, or inconsistency;
- trustworthy operation, by developing a certification framework for users, software, providers, and resources (especially IoT devices), and by properly linking identity and security requirements at the Userland layer with the specific identity management, access control, privacy models, encryption functions in the physical resources.

The starting point for this layer could be existing open-source software for building network operating systems, like ONOS [13]; in this respect, it is necessary to understand whether the ONOS Core is functionally and semantically rich enough for the purpose of virtualising a distributed data center environment, in addition to the transport network.

C. Userland layer

The Userland layer represents a middleware on top of the virtual resources exposed by the kernel, which

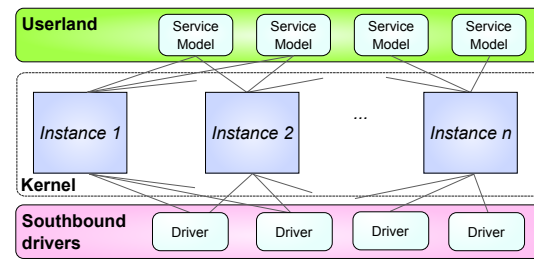


Fig. 4. The Kernel architecture for high-availability.

builds an enhanced execution environment for users' applications. Similarly to usual utilities and libraries in an OS, the the Userland exposes APIs and system calls that offer high-level abstractions of kernel resources, hence facilitating the development of services and applications. It should include different service models (e.g., Infrastructure as a Service – IaaS, Platform as a Service – PaaS, Software as a Service – SaaS), access to context information, public data sets, service platforms, third-party applications, an so forth.

The concept is essentially the same of the FIWARE platform [18], where a set of generic and specific enablers are available to ease the development of Smart Applications in multiple vertical sectors. Indeed, the FIWARE platform could be seen as a possible implementation of the Userland (or could be part of it), while the CIOS kernel is an enhancement of the current FIWARE-Lab platform (which currently is only an OpenStack installation with multiple regions) [19].

IV. RUNNING APPLICATIONS OVER THE CLOUD CONTINUUM

The concept of cloud continuum provides a great enhancement in the way applications are deployed and run with respect to current practice.

On the one hand, on-boarding and deployment of applications is a time-consuming and expensive task just for a single cloud, and the risk is that it become too cumbersome for distributed heterogeneous clouds [20]. As a matter of fact, interoperability and data/application portability are challenging issues for enabling rapid and seamless migration of software components and entire applications with minimal service disruption in case of failure or resource optimization [2]. Despite the effort in standardizing common interfaces (e.g., OCCI and CIMI⁴), there are currently many different open and proprietary APIs for Cloud Management, based on different abstractions, semantics and communication protocols. In this case, a common interface to the cloud continuum frees developers and operational staff from on-boarding applications on heterogeneous infrastructures, counting on the underlying Cloudscape for redundancy and locality management. On the other hand, the availability of

⁴Cloud Infrastructure Management Interface (CIMI), by the Distributed Management Task Force. URL: <https://www.dmtf.org/standards/cloud>.

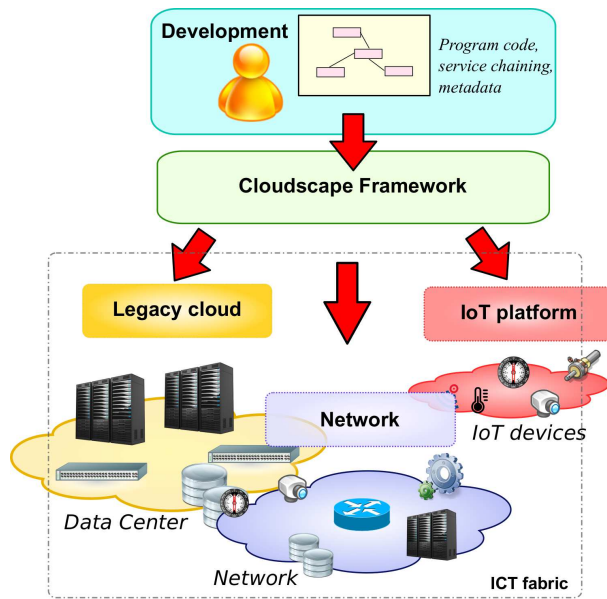


Fig. 5. Service orchestration over the CIOS.

heterogeneous resources at the very edge of the network as well as high performance computing islands enables users to include locality and distance policies in their applications. Such information could be exploited by the CIOS to reduce the amount of information to be transferred over the network, to shrink the latency and responsiveness of applications, to bring redundancy and resilience into the services, to fulfill privacy regulatory restrictions. In both cases, the full potential of this approach will only be effective if the development and deployment paradigms were able to reflect the underlying cloud capability, and to provide suitable software models and abstractions to programmers (see Fig. 5).

Finally, to exploit the full potential of the envisioned Cloudscape framework, current model-driven software development paradigms (e.g., TOSCA [21]) should be extended to include additional facets, concerning locality, distance, performance, Quality of Service (QoS), security, and dependability policies. This will allow the CIOS to optimize the behavior of an application by suitable orchestration and placement strategies.

V. CONCLUSIONS

In this paper, we have discussed our idea about a possible architecture to create cloud abstractions over an ICT fabric, in order to build the cloud continuum. We outlined the layout of a framework that follows the same structure of a typical modern Operating System, and explained how this structure is suitable to provide common functional abstractions of heterogeneous infrastructures (data centers, ICT fabrics, IoT), to guarantee scalability and availability, and to expose different service models.

Our work aims at proposing a novel approach for building large distributed cloud systems beyond mere federation and to stimulus the discussion on this topic.

Our next steps will be to investigate further the structure in the context of 5G network and upcoming protocols and technologies.

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